



THE TECHNICAL UNIVERSITY OF KENYA

FACULTY OF APPLIED SCIENCES AND TECHNOLOGY

SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

CONCEPT PAPER

PROJECT TITLE: BORESHA-MAMA: MOBILE AND WEB-BASED
MATERNAL HEALTHCARE SYSTEM FOR TRANS-NZOIA COUNTY

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**A PROPOSAL SUBMITTED TO THE SCHOOL OF COMPUTING AND
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1. OVERVIEW OF THE RESEARCH PROJECT

Maternal healthcare in rural Kenya, particularly in Trans-Nzoia County, is characterized by significant access challenges. Pregnant women often face long travel distances to reach health facilities, rely on fragile paper-based record systems, and incur prohibitive costs associated with frequent clinic visits. This results in missed antenatal and postnatal appointments, delayed detection of pregnancy complications, and fragmented care throughout the maternity journey.

Existing digital health solutions in Kenya tend to focus on urban settings or general healthcare, lacking specialized tools for the complete maternal care continuum in resource-limited, rural contexts. While programs like Linda Mama address financing and M-TIBA facilitates mobile payments, there remains an absence of an integrated, end-to-end digital platform that connects expectant mothers, community health volunteers (CHVs), local health facilities, and county health managers into a coordinated system.

This research project aims to design, develop, and evaluate **Boresha Mama**—a purpose-built mobile and web-based system for maternal healthcare. The platform will enable mothers to digitally track pregnancy milestones and receive care, empower CHVs with tools for efficient home-based follow-up, equip health facilities with dashboards for patient management, and provide county officials with data for oversight and planning. It seeks to replace unreliable paper trails with a secure, accessible digital record, thereby improving service coordination, timely intervention, and health outcomes for mothers and newborns in Trans-Nzoia.

2. STATEMENT OF THE RESEARCH PROBLEM

The core issue is the lack of a cohesive digital infrastructure to support maternal healthcare delivery in Trans-Nzoia County. This deficiency manifests in several interconnected problems:

1. **Disconnected Care Pathway:** Mothers, community health volunteers (CHVs), frontline health facilities, and county management operate in informational silos. Critical patient data captured during home visits does not flow seamlessly to clinics, and referral outcomes are rarely tracked, leading to disjointed care.
2. **Inefficient and Unreliable Data Management:** The reliance on paper-based records (Mwanaichi cards) and manual registers makes patient information vulnerable to loss, damage, and inaccessibility. This often results in repeated medical tests, incomplete health histories during emergencies, and inefficient use of clinical time.
3. **Barriers to Service Utilization:** Geographical distance (often 50–100 km to equipped facilities), combined with the direct costs of transport and the indirect cost of lost income, deters mothers from attending the recommended number of antenatal and postnatal check-ups, increasing risks for undetected complications.
4. **Operational Inefficiencies for Health Workers:** CHVs are burdened by cumbersome paper forms, which slows down data collection, risk assessment, and the initiation of referrals for high-risk cases. Health facility staff lack digital tools for scheduling, tracking patient progress, and generating reports, reducing their operational capacity.
5. **Limited Data for Decision-Making:** County health managers lack a real-time, consolidated view of key maternal health indicators across sub-counties. Planning and resource allocation are based on aggregated, delayed paper reports rather than actionable, timely data.

Primary Research Question: How can a user-centered, mobile-based digital system be effectively designed and implemented to bridge the information and coordination gaps in the maternal healthcare ecosystem of Trans-Nzoia County?

Hypothesis: The introduction of an integrated digital platform (Boresha Mama) that connects all stakeholders will lead to improved antenatal care attendance, more efficient CHV workflows, better-coordinated referrals, and enhanced capacity for data-driven health management at the county level.

3. IMPORTANCE OF THE RESEARCH PROJECT

This project is significant for its potential to create tangible impact across multiple domains:

- **Clinical and Public Health Impact:** By facilitating earlier and more consistent engagement with health services, the system can directly contribute to reducing preventable maternal and neonatal morbidity and mortality. It empowers the early identification and management of high-risk pregnancies.
- **Socio-Economic Impact:** The platform can alleviate the financial strain on households by reducing the frequency and necessity of long-distance travel for routine care. It also respects the time of mothers and health workers by making processes more efficient.
- **Operational and Health Systems Impact:** Boresha Mama aims to streamline workflows for CHVs and clinic staff, reducing administrative burdens and allowing them to focus more on patient care. It provides the county health department with a powerful tool for monitoring, evaluation, and evidence-based planning, aligning with Kenya's Universal Health Coverage (UHC) agenda.
- **Academic and Technological Contribution:** The project offers a practical case study in applying appropriate information technology to solve a pressing real-world problem in a low-resource setting. It will generate insights into the design principles, implementation challenges, and success factors for digital health interventions in rural Africa, contributing valuable knowledge to the fields of health informatics and community health.

The solution is designed for scalability and adaptation, offering a model that could be replicated or customized for other counties facing similar maternal health challenges.

4. PROPOSED SOLUTION

The **Boresha Mama** system is a multi-component platform designed to serve the distinct needs of each actor in the maternal care chain.

4.1 System Components & Users:

1. **Mother Mobile Application:** A smartphone app allowing pregnant women and new mothers to maintain a digital maternity record, receive personalized appointment reminders and health tips, track pregnancy progress, and access information about their nearest health facility.
2. **Community Health Volunteer (CHV) Mobile Application:** A field application for CHVs to register new pregnancies during home visits, log follow-up assessments, record vital signs, screen for danger signs, initiate electronic referrals to clinics, and sync data when connectivity is available.
3. **Health Facility Web Dashboard:** A browser-based dashboard for nurses, midwives, and clinical officers at dispensaries and health centers. It enables them to view and update patient records, manage appointments, track referred cases from CHVs, view pregnancy timelines, and generate basic clinic reports.
4. **County Health Management Web Portal:** A comprehensive dashboard for sub-county and county health managers to oversee the system. Functions include managing CHV and facility accounts, monitoring key performance indicators (e.g., ANC coverage, referral rates), visualizing data trends on maps and charts, and generating reports for higher-level decision-making.

4.2 Proposed Technology Stack:

- **Backend Server & API:** Node.js runtime with the Express.js web application framework. This will handle business logic, user authentication, and data processing.
- **Database:** PostgreSQL, a robust, open-source relational database system, chosen for its reliability, strong data integrity features, and suitability for handling structured healthcare data securely.
- **Frontend (Web):** React.js library for building dynamic and responsive user interfaces for the health facility and county management dashboards.
- **Frontend (Mobile):** React Native framework for developing the cross-platform mobile applications for mothers and CHVs, allowing a single codebase to run on both Android and iOS devices.

Supporting Tools & Services:

- **Version Control:** Git with GitHub/GitLab for collaborative code management.
- **API Development & Testing:** Postman.
- **SMS Gateway:** Integration with a local provider like Africa's Talking API for sending appointment reminders and alerts via SMS/USSD, ensuring reach to users with basic phones.
- **Development Environment:** Visual Studio Code.

4.3 Implementation Approach:

The development will follow an iterative approach, starting with core features for a pilot in one sub-county (e.g., Kiminini). Extensive user feedback will be incorporated throughout the design and testing phases. Key considerations include designing for offline functionality in the CHV app, ensuring a simple user interface for varying literacy levels, and adhering to data privacy and security standards as per Kenyan regulations.

My technical capabilities:

- Good at JavaScript (Node.js, React).
- Good with PostgreSQL database.
- Basic Python knowledge.

Yes, I can build this system with these skills

5. REFERENCES

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